

CEDAR
Healthcare

THE FUTURE OF
HEALTHCARE

Our
HealthCare
Product

www.ayninfotech.com

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1 Healthcare in Transformation

There has been unprecedented scientific progress in healthcare-related technologies, from medical health trackers to low-cost genome sequencing, telemedicine, and population health management tools. The data from these sources is analyzed and studied correctly by the healthcare users using cursory integration of data sources.

The convergence of these data sources, proper management of complex data, and application of Machine and Artificial Intelligence will usher a new era of healthcare. One in which healthcare solution helps manage patients' medical data is based on

their own unique medical, proteomic, genomic factors. Moreover, the healthcare system should also consider a person's socio-economic and social profile, i.e., medical history. It is a correct vision of a person-centric healthcare system.



2 Current Challenges in Healthcare Industry

2.1 Absence of Medical History

Despite all technologies, medical history is the mainstay of a patient's diagnosis. The medical history must include illness and the impact of hereditary, social, environmental, and behavioral factors on a patient's well-being. The medical interview of each patient is essential and can help in diagnosing the root disease.

A complete and exact history is the establishment of all future medical consultations.

The majority of people in India, while consulting with doctors are unable to share a proper medical history as most of them are unaware of its importance. People are even unaware of the past medical history they had and thus missed out sharing essential medical data with doctors. Few times due to lack of information, doctors are unable to give correct medication to patients, and in severe cases, it leads to fatality.

In 2018, a study at Harvard University shows that nearly 5,000,000 deaths occur in India due to medical error every year.

Reason to Collect Patient History

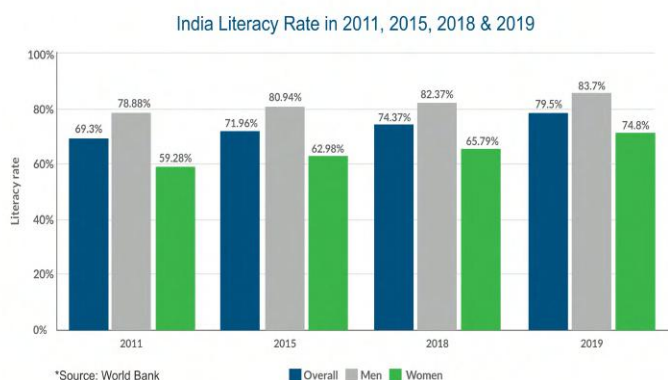
1. Family history is a traditional tool for diagnosis of hereditary diseases and identifying genetic disorders. It detects severe genetic disease like cancer, heart attack, diabetes, and asthma at an early stage.
2. Family history helps doctors to make better decisions about diagnosing diseases and prescribing proper medicines to patients accurately.
3. Family history helps physicians build rapport between patients' families and understand the patient's environment and behavior.
4. It helps identify the inheritance and pattern of disease in the family, for instance ,a family has a disease after each generation.



2.2. Lack of Awareness and Education

Education is a basic necessity of society. Education empowers people to learn, understand, and create new possibilities. Due to the lack of knowledge and unawareness about health science, people tend to hide medical symptoms and hereditary diseases.

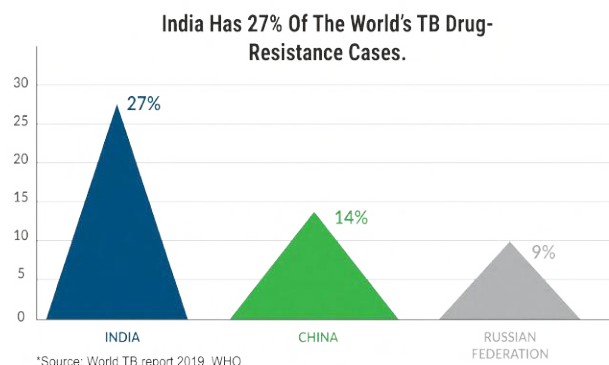
Proper education spreads awareness regarding health science and makes people understand the seriousness of any diseases. The below chart is showing the literacy rate of India from 2011 to 2019. Even though the literacy rate increases year by year, millions of people are unqualified and lack understanding.



2.3. Drug-Drug Interaction (DDI)

Medicines help cure, halt and prevent diseases. However, due to the nescience of current medication and patient's medical history, doctors prescribe medicines which further lead to adverse-effect of drug-drug interactions. Avoiding food habits during medication leads to the side effect of drug-food interaction. Below bar graph shows drug resistance cases for tuberculosis in India.

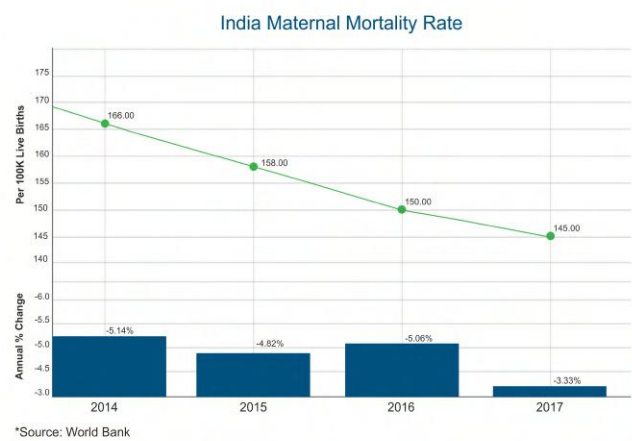
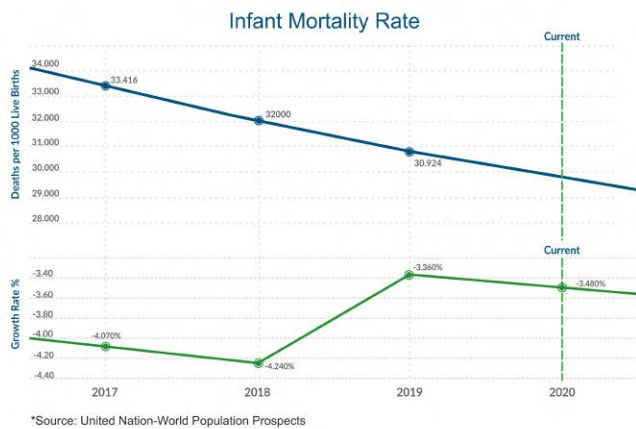
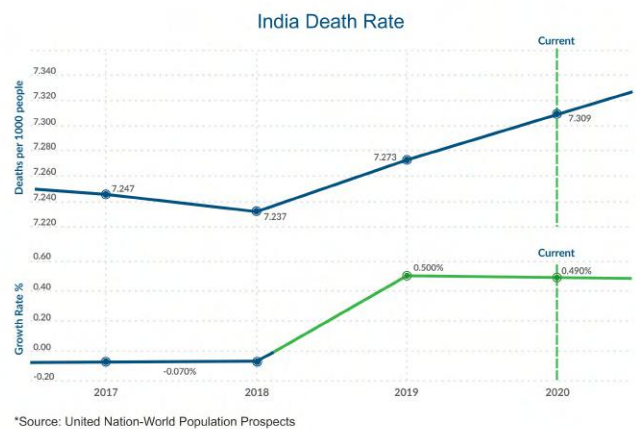
In 2018, there were about half a million new drug resistance TB cases.



2.4. Mortality Rate

The world is facing significant issues on a rural scale for healthcare facilities. As a substantial population lives in rural areas, healthcare facilities need to reach villages. Due to a lack of medical facility, awareness, and education, the mortality rate is growing. Infant mortality and the maternal mortality rate is the top concern for medical professionals.

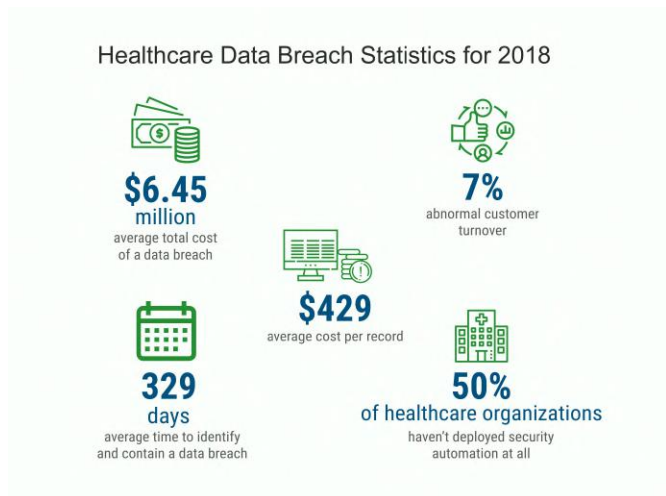
The below graphs show the data on India's mortality rate.



2.5 Security Risk to Patient Data

Rural Health Care in India manages patient medical data in a traditional paper-based structure, while urban healthcare is operating data digitally, mostly on the cloud.

Healthcare users manage Electronic Health Record (EHR) in a centralized database system where most of the medical information is portable, easily accessible, and highly insecure.



2.6. Insurance Fraud

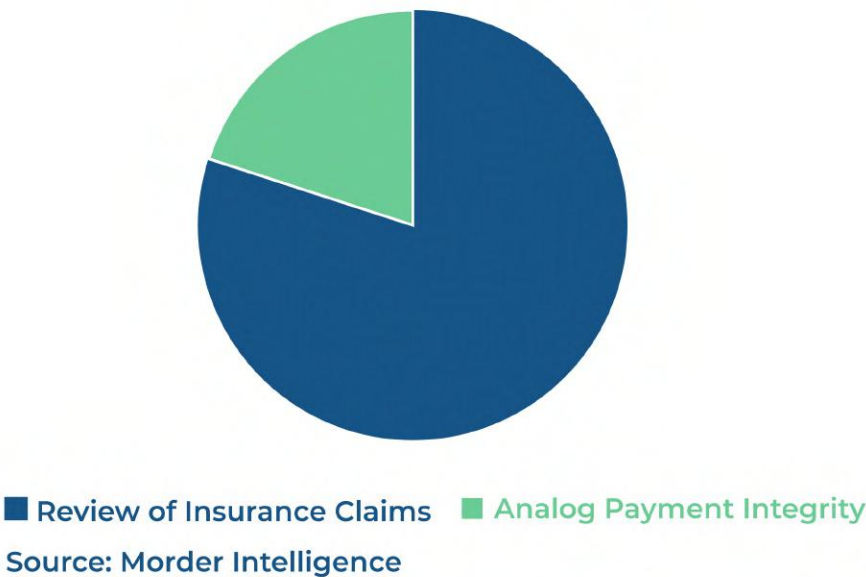
The insurance industry is proliferating due to COVID-19. Due to immense growth, it leads to higher chances of insurance fraud, like healthcare users who enter their false diagnosis condition in health records to get an insurance claim.

Moreover, people are unaware of the insurance policies, and most of it remains unclaimed at times.

In 2018, the Insurance company loses over the US \$6.25 billion to fraud, resulting in a higher premium for genuine customers.



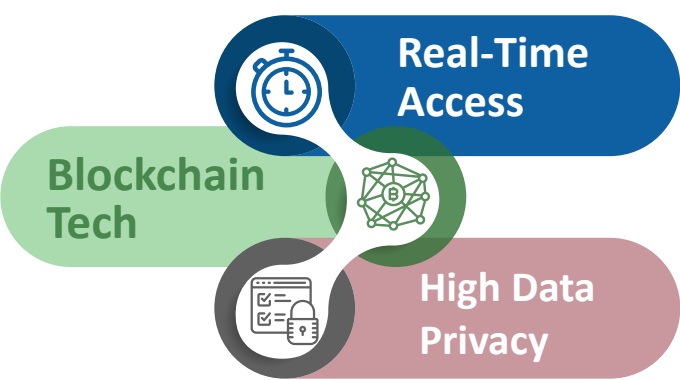
Healthcare Fraud Detection Market: Revenue Share (%),
By Application, Global, 2018



3 Cedar Healthcare: Solution for Today's Challenges

Cedar Healthcare empowers doctors and healthcare organizations to transfer and access protected healthcare information securely. It creates a trusted data and information ecosystem for pharma and doctors to collaborate, allowing each of them to access the data they provide for more effective patient care. As data privacy is the primary factor, Cedar Healthcare can summarize and present only the critical data or agreed patient information. Cedar healthcare eliminates the current healthcare environment's challenges by adapting to the changing healthcare sector trends.

There are three main features of Cedar Healthcare:



3.1 Real - Time

Accessing data is vital to deliver services on new scientific advances. Cedar Healthcare provides users with instant access to health records. Users can instantly access records anywhere and at any time.

3.2 Blockchain Technology

Cedar Healthcare leverages Blockchain technology, an open and secure technology that captures transaction records on blocks connected and stored on a distributed ledger. It has incredible security benefits and impossible to manipulate without permission. All the transactions that happen on Blockchain are dated and time-stamped.



Cedar healthcare is purposely built on Blockchain Technology.

Key Capabilities of Blockchain are:

a. Reliable on-Chain Data Management

Blockchain is the only commercially available platform that combines on-chain big data, immutability, HIPAA, NIST, and GDPR compliances.

b. Multidimensional Health Profile

Blockchain helps build various health profiles, including people, places, and things, and helps summarize all types of data like insurance claims, pharmacy reports, and every data can be managed easily with Blockchain.

c. Globally Accessible

Blockchain enables us to securely share data with partners, healthcare users, and communities across the globe to accelerate research and innovation without losing control over data.

d. Data Ownership

Blockchain allows users to control everything about how their data is shared. Every update regarding the information is updated with the data owner.

e. Development Accelerators

Blockchain helps users get into the market faster, and it enhances its solution's capabilities using a full suite platform and application tools: dynamic schemas, data mapping libraries, orchestration tools, and app framework.

3.3 High Data Privacy

Cedar healthcare works with HIPAA- a complaint protected health information storage that adheres to region-specific regulatory guidelines. HIPAA (Health Insurance Portability and Accountability Act of 1996), provides security provision and data privacy, to keep patient health records secure and confidential.



4 Benefits of Cedar Healthcare

4.1 Proper Data Management

Patients currently have healthcare records at multiple hospitals, networks, and different cities or countries. There are numerous fragmented records of a single patient at pathological labs, hospitals, and pharmacists and medical reports, clinic letters, and other bills that are improperly arranged.

Cedar Healthcare arranges all the data and reports in chronological order and filters them properly in the specific category. Such categorization would make data more accessible and easy to handle. Healthcare users will easily access the data they need and will be a big time saver.

Cedar Healthcare empowers users to securely access, track and store data and exchange health information with the chosen medical professional.

4.2 Security of Users

The Cedar system uses Blockchain technology to maintain and secure the data. It is mainly based on a closed, permission-based Blockchain. As Blockchain is a distributed ledger, the data is inaccessible and impossible to manipulate, as well as to hack data.

The Blockchain acts as a pointer, where medical data is stored in an encrypted form, which means that if anyone tries to access data, they will be unable to do so.



4.3 Better Digital Communication

Cedar Healthcare centralizes all the medical data to manage, share, and track data easily. Whenever any authorized user changes the health record, the system will update the record on the system. Cedar Healthcare is accessible in any system with an internet connection and a computer or a

4.4 Borderless Sharing of Data

Blockchain technology provides a platform accessible to authorized users only. People who frequently travel around the globe can have their health records along with them across borders. Healthcare service becomes borderless by using digital technologies like telemedicine, which helps doctors and patients to connect.



4.5 Health Insurance Benefit

Patients knowingly or unknowingly avoid disclosing their health condition to the

insurance company. Due to unawareness, they are unable to take benefits of health insurance. Due to incorrect information, the patient pays the incorrect premium for years and discovers that the company declines their claims.

The verifiability and immutability of the health record on the Cedar platform allow insurance companies to verify and access information properly. They make a proper assessment of the patient's health condition, and health premiums should reflect this.



5 About AYN InfoTech

AYN InfoTech started in 2009 with a vision to provide world-class IT services and smart products across the world. We registered AYN InfoTech in the year 2017 and achieved great success by satisfying our esteemed customers.

We are a fast-growing IT company in Pune, India; we have grown our business and captured market trust and satisfaction. We have a team of technology experts to understand client's requirements and provide solutions to compete with other markets.

Our team of specialists consistently delivers outstanding results combining creative ideas with our vast experience. We can help you build a sustainable, meaningful relationship with your clients by engaging them with your brand using Social Media.



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For More Information, Visit:
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